MIL-M-21383A
22 October 1976
Superseding
MIL-M-21383(NOrd)
28 April 1958

MILITARY SPECIFICATION

MANGANESE DELAY COMPOSITION

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification governs the preparation of manganese delay composition and the methods of inspection and tests upon which product acceptance will be based.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposals form a part of this specification to the extent specified herein. In the event of conflict between this specification and other documents referenced herein, requirements of this specification shall apply.

SPECIFICATIONS

Federal

PPP-B-621	Boxes, Wood, Nailed and Lock Corner
PPP-C-96	Can, Metal, 28 Gage and Lighter

Military

JAN-M-476	Manganese, Powdered (For Use in Ammunition)
JAN-L-488	Lead Chromate (For Use In Ammunition)
MIL-B-550	Barium Chromate
MIL-C-10464	Cans, Hermetic Sealing, Metal Light Gage Tear-Strip Type
MIL-P-22264	Powders, Ignition, Gasless
MIL-V-23776	Vermiculite, Expanded
MIL-B-82647	Bag, Conductive, Heat-Sealable, Plastic

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Ordnance Station, Standardization Division (611), Indian Head, MD 20640, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

Military

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-1234	Pryotechnics: Sampling, Inspection, and Testing

DRAWINGS

Naval Sea Systems Command (Code Ident. 10001)

Dwg 423038	Replacement Sample Powder Tank Mk 1 and Mods
LD 480089	Test Element and Output Charge Holder
Dwg 1826685	Delay Body
Dwg 1826687	Test Element No. 3, Loading Procedures
Dwg 1826740	Output Charge Holder, Loading Procedures

PUBLICATIONS

Naval Sea Systems Command (Code Ident. 10001)

OD 9360	Preparation of Manganese Delay Composition

Naval Air Systems Command

AD 1014	Operating Procedures for Pyrotechnic Delay Timing Apparatus

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 *Formulation of manganese delay compositions.* The manganese delay composition shall be formulated from treated manganese powder, and barium chromate and/or lead chromate. Advisory methods for preparing, proportioning, and compounding ingredients to obtain designated burning time values are provided in OD 9360.

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3.1.1 Manganese powder.

3.1.1.1 Untreated manganese powder shall conform with JAN-M-476, Grade I.

3.1.1.2 The particle size of untreated manganese powder shall be within the range of 10 to 14 microns weight average diameter.

3.1.1.3 The average burning time of the ambient shots of manganese powder treated with Anstac solution shall be less than 3.5 seconds.

3.1.1.4 The average burning time of the environmental shots of manganese powder treated with Anstac solution shall not differ more than 15 percent from that of the ambient shots.

3.1.2 Barium chromate.

3.1.2.1 Barium chromate shall conform to MIL-B-550, Grade A.

3.1.2.2 The average particle size of the barium chromate shall be 2 microns or less.

3.1.3 Lead chromate.

3.1.3.1 Lead chromate shall conform to JAN-L-488.

3.1.3.2 The average particle size of the lead chromate shall be 2 microns or less.

3.2 Burning time. The burning time of manganese delay composition shall conform to the basic time and tolerance designated in the drawing, specification, or contract, as furnished. The burning time is an average time ascertained under the provisions of 4.3.4. The percentage of coefficient of variation of observed burning times shall be less than 3.0 as determined by 4.3.4.4.

3.2.1 Shipment or prolonged storage. After manganese delay powder has been properly mixed, it is possible for the fuel and oxidants to segregate under certain conditions of storage and transportation. Segregation appears to occur near the surface when a delay composition is subjected to prolonged agitation, particularly during shipment by rail. The surface layer is then nonhomogeneous and not representative of the composition of the lot. Therefore, after shipment of a lot of manganese delay composition or after storage of the lot for longer than 3 months, the lot should be reevaluated to the requirements of 3.2.

3.3 Environmental requirement. The burning of the composition shall not change by more than 15.0 percent after exposure for 120 hours to an atmosphere of 95 ± 1.0 percent relative humidity and at a temperature of 71° ± 1°C. The provisions of 4.3.5 shall be applicable in testing and ascertaining conformance.

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3.4 Low temperature functioning. The burning time shall not be increased by more than 15.0 percent when the initial temperature is changed from 24° to -54°C. The provisions of 4.3.6 shall be applicable in ascertaining conformance.

3.5 Delay performance. There shall be no delay failures at room temperature, at low temperature, or after environmental testing. A delay failure shall be a condition where ignition powder ignites but a delay column does not burn through. Satisfactory performance in accordance with 4.3 with no delay failures shall constitute conformance to this requirement.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may utilize his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specifications where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Lot. A lot of manganese delay composition shall consist either of a manufacturer's batch which has been mixed as a unit from the ingredients or of a blend of one or more such batches.

4.3 Tests.

4.3.1 Untreated manganese powder. The following procedure shall be followed for sampling untreated manganese powder for particle size analysis:

(a) Divide the lot into portions of less than 100 pounds.

(b) Quarter each portion and sample each quarter.

(c) Test each sample on the Micromerograph.

(d) If a sample fails to meet the requirements, screen the quarter portion through a 230-mesh sieve and retest the fines.

4.3.2 Treated manganese powder. The samples shall be treated with Anstac solution as follows:

(a) Mix Anstac solution with manganese powder in a ratio of 1 milliliter of solution to 1 gram of powder.

(b) Evaporate solution in air and dry powder 10 minutes at 71°C.

(c) Screen treated manganese powder through a 230-mesh or finer sieve.

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(d) Mix 45 grams of treated manganese powder with 55 grams of lead chromate for 30 minutes in a 1/2-pint ball-mill jar containing about 15 1/2-inch porcelain balls.

(e) Load 10 test elements per 4.3.1.1. Test 5 at 24 ± 4°C per 4.3.4.2 and 5 after environmental conditioning as described in 4.3.5.

4.3.3 Particle size. The average particle size of barium chromate and lead chromate shall be determined by the procedure of MIL-STD-1234, method 202.1.

4.3.4 Burning time test. The test sample shall consist of 20 test elements as specified in 4.3.4.1. The lot as specified in 4.2 shall be divided into 20 equal parts, and 20 delay bodies shall be loaded so that each delay body contains a representative sample from one of the 20 parts without duplication.

4.3.4.1 Loading of test element. The test element shown in Dwg 1826685 shall be loaded according to directions on Dwg 1826687. The tools required for loading appear on drawings under LD 480089. The ignition increment, P1, of approximately 105 milligrams of A1A composition in accordance with MIL-P-22264 shall be adjusted to be within the drawing limits for increment length when pressed at 30,000 ± 500 psi. The delay composition increments shall be charged separately at 30,000 ± 500 psi to be within the drawing limits for total length of delay column. Each increment may be measured volumetrically by means of a scoop of such size as to give six approximately equal increments. After the last increment, D6, is pressed, the loading funnel cap shall be loosened and tightened again. The D6 increment shall then be repressed to improve compaction and to minimize the danger of loss of powder by fallout. The surface of the delay column shall be flush with the end of the delay element in which it is contained. Excess powder may be removed by sliding the end of the test element over a hard cloth. The output charge holder containing the P2 charge shall be loaded in accordance with Dwg 1826740.

4.3.4.2 Timing of test elements. Directions for mounting components into the delay body holder and assembling the test adapter are given in AD 1014. The temperature of the test elements for room temperature firing shall be 24° ± 4°C. If detailed directions for testing at low and high temperatures are not specified for the qualification of the powder, the following procedure shall be followed: Test adapters shall be conditioned continuously for at least 2 hours before firing at the required temperature.

4.3.4.3 Procedure for operating test apparatus. It is suggested that AD 1014 be used for operating instructions of test apparatus.

4.3.4.4 Computation of burning time and percentage of coefficient of variation. Compute the burning time which is the average ($\bar{X}$) of the time observations obtained in 4.3.4.2. Compute the standard deviation, (s), of the time observations. The symbols and the method of computation shall be governed by 6.3. Compute the percentage of coefficient of variation by dividing (100) times (s) by $\bar{X}$. Disposition of the lot shall be governed by 3.2 and 3.5.

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4.3.4.5 Testing after shipment or prolonged storage. As specified in 3.2.1, lots shall be tested as specified in 4.3.1. Those lots failing to meet the requirements of 3.2 shall be remixed to restore uniformity prior to further testing and use.

4.3.5 Environmental test. Sampling shall be conducted after the last mixing operation. Select and prepare a representative pooled sample of delay mixture from the lot. Prepare 10 test elements as specified in 4.3.4.1. Expose the test elements to an atmosphere of 95 ± 1.0 percent relative humidity for a period of 120 hours at a temperature of 71° ± 1°C. The exposure chamber shall be a desiccator or other sealed chamber where exposure to vapor in equilibrum with saturated potassium sulfate solution is obtained. True saturation of the solution shall be maintained by the presence of undissolved solute in contact with the solution. After exposure, dry the specimens in a ventilated oven for 4 hours at 71° ± 1°C. Condition the specimens for at least 4 hours in an atmosphere of 40 to 50 percent relative humidity and at a temperature of 24° ± 6°C. Ascertain the burning time of the surveillance sample by the procedures of 4.3.1.2 and 4.3.4.4. Compute the percentage of change in burning time on the basis of change from the burning time ascertained under 4.3.4. Disposition of the lot shall be in accordance with 3.3 and 3.5. A lot rejected for not conforming to 3.5 may be remixed and resubmitted for acceptance.

4.3.6 Low temperature functioning test. Sampling shall be conducted after the last mixing operation. Select and prepare a representative pooled sample of delay mixture from the lot. Prepare 10 test elements as specified in 4.3.4.1. Ascertain the burning time at low temperature by the procedures of 4.3.4.2 and 4.3.1.4 with the exception that the temperature at the time of firing and for at least 4.0 hours prior thereto shall be -54° ± 2°C. Compute the percentage of change in burning time on the basis of change from the burning time ascertained under 4.3.4. The lot is acceptable if the increase in burning time is less than 15.0 percent as required by 3.4 and if the lot meets the requirement of 3.5. A lot rejected for not conforming to 3.5 may be remixed and resubmitted for acceptance under 4.3. Failure of the lot to confrom to 3.5 on resubmission shall result in rejection of the lot.

5. PACKAGING

5.1 Packaging.

5.1.1 Inner container. Place approximately 1,200 grams of manganese delay composition into a doubled bag of four mil conductive plastic formed by inserting a 6 x 12-inch bag into another. The plastic bags shall conform with MIL-B-82647. Twist the top of the bag tight, fold over once, and secure with two No. 84 rubber bands.

5.1.2 Intermediate container. Place two doubled bags in a metal ammunition can conforming to Drawing 423038. Expanded vermiculite material conforming to MIL-V-23776 shall be used inside the intermediate container to ensure a tight pack. Maximum weight per can is about 5 lbs. or:

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5.1.2.1 Intermediate container, alternate #1. Place two doubled bags in a can conforming to MIL-C-10464 and seal according to conventional canning techniques. This can is hermetically sealed. Expanded vermiculite material conforming to MIL-V-23776 shall be used inside the intermediate container to ensure a tight pack. Maximum weight per can is about 5 lbs.

5.1.2.2 Intermediate container, alternate #2. Place two doubled bags in a one gallon capacity, friction top container conforming to PPP-C-96, Type V, Class 2. The cover shall be fixed by crimping at four or more points. Expanded vermiculite material conforming to MIL-V-23776 shall be used inside the intermediate container to ensure a tight pack. Maximum net weight per container is about 5 lbs. manganese delay composition.

5.2 Packing. Intermediate containers shall be packed in a domestic type, wood-cleated plywood box or in a Class 1, Style 2, 2-1/2 3, or 4 nailed wooden box conforming to PPP-B-621. Vermiculite material per MIL-V-23766 shall be used within the shipping box to provide separation of the intermediate containers and to ensure a tight pack. The gross weight of a given shipping container shall not exceed 50 lbs.

5.3 Marking. In addition to the markings required by contract or order, unit packages and shipping containers shall be marked in accordance with the requirements of MIL-STD-129. In addition, the label on each container shall bear the following information:

Manganese delay composition
(Manufacturer, date of manufacture)
(Lot number)
(Burning time at room temperature)
(Coefficient of variation)

5.4 Data cards. Three data cards, 5 by 8 inches, shall be furnished with each lot; one to the procuring activity, one to the consignee, and one to be packed in each container. Each data card shall have the following information;

Manganese Delay Composition

Manufacturer _____
Manufacturer's lot no. _____
Burning time (average):
(a) 24 ± 4°C _____
(b) -54°C _____
(c) After environmental test _____
Standard deviation
(a) 24 ± 4°C _____
(b) -54°C _____
(c) After environmental test _____

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Manganese Powder

Manufacturer _____
Manufacturer's lot no. _____
Surface treated by _____
Date of treatment _____

Barium Chromate

Manufacturer _____
Manufacturer's lot no. _____

Lead Chromate

Manufacturer _____
Manufacturer's lot no. _____

6. NOTES AND CONCLUDING MATERIAL

6.1 Intended use. Manganese delay composition is intended for use as a delay time determining element in ignition devises such as fuze explosive trains and cartridge actuated devices. The mixture burns substantially without evolution of gas; however, venting or expansion chamber pressure relief is desirable to relieve pressure generated by ignition elements. Application of this powder is restricted to hermetically sealed assemblies.

6.2 Ordering data. Procurement documents should specify the following:

6.2.1 Procurement requirements

(a) Title, number, and date of this specification

(b) Burning time (3.2)

(c) Inner and outer container (5.1, 5.2)

(d) Marking (5.3)

(e) That the safety precaution requirements of the Contractors' Safety Manual for Ammunition, Explosives and Related Dangerous Material, DOD 4145.26M, are applicable. NOTE: When this specification is used as part of the description of work to be accomplished by a Government activity, the safety precaution requirements of Ammuntion Ashore, OP 5, should be made applicable.

6.2.2 Contract data requirements. Items of deliverable data required by this specification are cited in the following paragraphs herein:

Paragraph	Data Requirement	Applicable DID
5.4	Data cards	*

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DIDS (Data Item Descriptions/DD Forms 1664) for the above data requirements are/will be (*) documented in the applicable ADL (Authorized Data List).

Such data will be delivered as identified on completed (numbered) DIDS when specified on DD Forms 1423 (Contract Data Requirements List) and incorporated into applicable contracts.

6.3 Computation of mean and standard deviation of a sample.

6.3.1 Mean: the sum of individual measurements divided by the number of measurements

$$\bar{X} = \frac{\Sigma X}{N}$$

6.3.2 Standard deviation:

$$s = \sqrt{\frac{N\Sigma X^2 - (\Sigma X)^2}{N(N-1)}}$$

where N = the number of measurements
ΣX^2 = sum of the squared measurements
$(\Sigma X)^2$ = sum of the measurements squared

6.4 Micromerograph Type XC and Anstac solution are available from Val-Dell Co., Norristown, PA.

6.5 The Occupational Safety and Health Administration recommends that worker exposure to hexavalent chromium be controlled or eliminated. Barium chromate is an inferred carcinogen and lead chromate is an evident carcinogen.

6.6 Due to the comprehensive nature of this revision the margins of this specification have not been marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of relationship to the last previous issue.

Custodians:
Navy - OS
Army - EA
Review Activity:
Army - PA

Preparing Activity:
Navy - OS
Project No. 1377-0610

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OMB Approval
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DOCUMENT IDENTIFIER AND TITLE

MIL-M-21383A Manganese Delay Composition

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

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